

Just a Number: Do AI Benchmarks Really Matter?

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Abstract

Every new AI model is launched with a table of benchmark scores, and these numbers shape headlines, investment and what people believe AI can do. This paper argues that **benchmarks are just a number**: a narrow, often misleading snapshot of one model, on one test, under one setup. We show how benchmarks are gamed, leaked into training data, saturated, and dependent on who runs them and how. We show that high scores often fail to predict real-world usefulness, drawing on public examples from 2026 and on our own experience training and evaluating models at Grow Labs. We also give the strongest arguments for benchmarks and accept what they do well. Our conclusion is that benchmarks are useful as a rough signal of progress, but they should never be treated as proof of intelligence or of value. What matters is how a model performs on the real task, for real users, over time.

1. Introduction

In September 2026, OpenAI released GPT-6 Astra with a headline score of 99.9% on ARC-AGI-3, a test built to measure general reasoning. When the ARC Prize team ran the same model on the same test with a neutral setup, it scored 62.7%. The model was the same. The number was not.

This single example captures the argument of this paper. A benchmark score is not a property of a model. It is the result of a model, a test, a setup, a scoring method and often a company's choices about what to report. Change any of these and the number changes.

Our position is simple:

"A benchmark is just a number. It tells you how a model did on a test. It does not tell you whether the model is intelligent, whether it is useful, or whether you should trust it."

This paper asks four questions:

- 1 What are AI benchmarks, and why did they become so important?
- 2 Why are benchmark numbers often misleading?
- 3 What are benchmarks actually good for?
- 4 What should replace a single number?

2. What benchmarks are

A benchmark is a fixed set of questions or tasks with known correct answers. A model is run on the set, and the result is a score, usually a percentage.

Benchmark type	Example	What it claims to measure
Knowledge	MMLU, GPQA Diamond	Facts and expert knowledge
Maths	FrontierMath	Advanced mathematical reasoning
Coding	SWE-bench, Terminal-Bench	Writing and fixing real software

Benchmark type	Example	What it claims to measure
Reasoning puzzles	ARC-AGI	General, flexible problem-solving
Computer use	OSWorld	Operating a real computer
Human preference	Chatbot Arena	Which answers people prefer

Benchmarks became important because they are fast, cheap and easy to compare. One number fits in a headline, a press release and an investor slide. That convenience is also the problem.

3. Why benchmarks are just a number

3.1 Goodhart's law: when a measure becomes a target

The economist Charles Goodhart observed that when a measure becomes a target, it stops being a good measure. AI companies compete publicly on benchmark tables. Every lab therefore has a strong incentive to improve those specific scores, whether or not the improvement reflects real ability. Models can be tuned, prompted and tested in ways that raise the number without making the model more useful.

3.2 The setup decides the score

The ARC-AGI-3 case is not unusual. Scores depend on the prompt format, the number of attempts allowed, the tools available, how long the model can think and which version of the test is used. Comparison sites reporting on Claude Fable 5.1 and GPT-6 Astra in September 2026 gave different scores for the same model on the same benchmark, for example 67.4% and 73.7% for Fable 5.1 on DeepSWE, because of different test setups. When the setup changes the result by several points, a lead of one or two points means very little.

3.3 Contamination: the answers may already be in the training data

Large models are trained on huge amounts of text from the internet. Many benchmark questions and answers are also on the internet. If a model saw the test during training, a high score may reflect memory, not reasoning. It is often impossible for outsiders to check, because companies rarely publish full training data.

3.4 Saturation: the tests run out of room

Benchmarks are designed to be hard, but models improve quickly. Tests that were once extremely difficult are now close to their maximum. GPT-6 Astra reportedly scored 97.6% on FrontierMath Tier 4 and 96.0% on GPQA Diamond. When the best models all score near the top, the benchmark can no longer tell them apart, and small differences are mostly noise.

3.5 Companies choose what to show

Launch announcements show the benchmarks a model does well on. Weaker results may be left out. In September 2026, GPT-6 Astra led Claude Fable 5.1 on maths and computer use, while Fable 5.1 led on Humanity's Last Exam and on an independent intelligence index. Depending on which rows you read, either model looks like the winner. Many scores are also reported by the company that built the model, not by an independent tester.

3.6 Leaderboards can be gamed

Even human-preference leaderboards are not immune. Researchers have raised concerns that companies can test many private versions of a model on public leaderboards and publish only the best result. People also tend to prefer longer, more confident and better-formatted answers, which is not the same as preferring correct ones.

3.7 AI ability is "jagged"

A single score hides uneven ability. Experts describe frontier AI as jagged: brilliant at some tasks and surprisingly weak at others that look similar to humans. A model can solve research-level maths and still make a basic mistake in a simple real-world task. An average across a test cannot show this.

4. High scores do not always mean real-world value

The most important problem is the gap between the test and real life.

What benchmarks measure	What real use requires
Short, well-defined questions	Long, messy tasks with unclear goals
One correct answer	Judgement, trade-offs and several acceptable answers
English and popular domains	Many languages and specialised fields
A single attempt, in a lab	Reliability over thousands of uses
Capability	Capability plus safety, cost, speed and trust

Public research has shown this gap. In a 2025 study by the research group METR, experienced open-source developers using AI coding tools on their own projects took about 19% longer to finish tasks, even though they expected AI to make them faster. At the same time, the same tools were posting strong results on coding benchmarks. The benchmark and the real workplace told different stories.

For communities like Romani speakers, the gap is even larger. Almost no benchmark exists for the Romani language. A model can top every leaderboard and still be close to useless for millions of people.

5. Our own experience: numbers that lied

At Grow Labs we have seen this problem directly, in our own models.

The overfitting gap. While training a small language model on an Apple M1 Max, we tracked the gap between validation loss and training loss. Increasing our dataset from 5,000 to 10,000 samples improved this gap from 0.5788 to 0.1851, an excellent number by the usual standard. But the model could still hallucinate or fall into repeating loops when it wrote text. The metric said the model was good. Reading its output said it was not ready. Only fixing the learning rate, dropout, weight decay and temperature addressed the real behaviour.

The Romani fine-tune. When we fine-tuned Mistral-7B on Romani-language books, the training loss went down as expected. But the loss curve could not tell us whether the model understood Romani better, had forgotten general knowledge, or had started inventing answers. No public benchmark could tell us either.

These experiences led us to build **Auditor**, a tool that does not rely on a fixed benchmark. It writes new test questions for each run, based on what the model was actually tuned for. About one question in four is a trap, such as a question about an event that never happened, to test whether the model admits it does not know. A separate judge scores each answer against criteria written before the answer is seen. Auditor also compares each run with earlier runs to catch real drops in quality. Even so, we treat its scores as a relative signal, not as truth. A number from our own tool is still just a number.

6. The case for benchmarks

A fair paper must take the other side seriously. Benchmarks do have real value:

- **They track progress over time.** Without shared tests, it would be hard to show that AI has improved at all.
- **They allow rough comparison.** A model scoring 30% and one scoring 90% on the same test really are different.
- **They catch regressions.** If a new version scores much lower, something has probably gone wrong.
- **They are reproducible.** Anyone can run the same test, which keeps claims somewhat honest.
- **They drive research.** Hard benchmarks such as ARC-AGI have pushed researchers to work on real weaknesses.
- **They can measure safety.** Tests for dangerous capabilities, such as cyberattacks, help labs decide when to add safeguards.

We accept all of these points. Our argument is not that benchmarks are useless. It is that **a benchmark score is evidence, not a verdict**. The problem is not the number itself. The problem is treating the number as the answer.

7. Better ways to judge AI

Instead of	Do this
One headline score	A profile of strengths and weaknesses across many tasks
Fixed public test sets	Fresh, private or newly generated tests that cannot be memorised
Company-reported numbers	Independent testing with published setups
General benchmarks only	Tests built for the real task, language and users
Single attempts	Reliability over many runs, including how often it fails
Accuracy alone	Accuracy plus honesty, safety, cost and speed
A one-time evaluation	Continuous monitoring after deployment
Only numbers	Reading real outputs and asking real users

The simplest rule is also the most important: **always read what the model actually produces**. No number replaces looking at the answers.

8. Discussion

If benchmarks are just numbers, why does the industry depend on them so much? Because numbers are simple, and AI is not. A benchmark gives investors, journalists and the public a way to rank something they cannot easily judge themselves.

This matters beyond marketing. Governments use capability scores to decide which models need regulation. Businesses use them to choose which model to buy. Researchers use them to claim progress towards AGI. If the numbers are misleading, all of these decisions are built on weak ground.

There is also a deeper point. Benchmarks measure what a model *can do* on a test. They say little about whether a model is *trustworthy*. As models become more capable and more autonomous, trust, honesty and control matter more than one more percentage point on a maths exam.

9. Conclusion

Benchmarks helped the AI field grow, and they still have a place. They are useful for seeing broad progress, spotting large differences and catching regressions. But a benchmark score is just a number. It depends on the test, the setup and the person reporting it. It can be gamed, memorised and saturated. And it often fails to predict whether a model will help a real person with a real task.

The right question is not "what did the model score?" It is "does the model work, reliably and safely, for the people who need it?" Answering that requires testing on real tasks, in real languages, over time, and above all, reading what the model actually says.